FAITH OGUNDIMU

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PROFESSIONAL SUMMARY

Computational cancer-genomics PhD researcher building machine learning for the non-coding genome, with first-author work in revisions at npj Genomic Medicine (Nature portfolio). Hands-on deep learning across convolutional, feed-forward and multimodal LLM systems, applied to chromatin accessibility, cfDNA methylation and single-cell multi-omics at the scale of thousands of tumour genomes. Turns large, messy biological data into models, rigorous evaluation and candidates for experimental validation; Government of Ireland Scholar (18% success rate).

EDUCATION

Royal College of Surgeons Ireland (RCSI) Sep 2025 - Aug 2029
Doctor of Philosophy (PhD) in Cancer Genomics and Epigenomics, Machine Learning and Bioinformatics
Thesis: Identification of non-coding somatic driver mutations in breast cancer

Dublin City University (DCU) Sep 2021 - May 2025
BSc. in Genetics and Cell Biology (GCB) | First Class Honours (1.1) | Ranked 2nd & Dean's Honour List
Final Dissertation: *Using neural networks and foundation models to understand gene regulation in the MCF-7 breast cancer cell line*

RESEARCH EXPERIENCE

Royal College of Surgeons Ireland (RCSI) Sep 2025 - Aug 2029
Doctoral Researcher
- Building ML (Hidden Markov Models, CNNs, MLPs) to prioritise non-coding driver mutations from >5,000 breast-cancer whole genomes (TCGA, ICGC, HMF), with CRISPR-based validation in cell lines.
- Constructing 3D tissue-specific regulatory maps and integrating single-cell ATAC-seq with ENCODE elements to target the 98% non-coding genome where drivers remain uncharacterised.

Beaumont RCSI Cancer Centre, Irish Cancer Society, Lung Health Check January 2026 - March 2026
Bioinformatics Research Assistant – BRAND Study (NSCLC Liquid Biopsy & cfDNA Methylation Analysis)
- Built an end-to-end cfDNA methylation and DMR-discovery pipeline across 88 bisulfite-sequenced plasma samples (~8.7M CpGs), reducing >89,000 candidate markers to 100 high-information CpGs (mean $|\Delta\beta| = 0.57$).
- Estimated tumour methylation fraction by semi-reference-free deconvolution for Ireland's largest lung-cancer pilot (€4.9M); found a significant post-immunotherapy TMF drop (Wilcoxon $p=0.015$, $r=0.85$) and ΔTMF as a near-significant survival predictor (Cox $HR=1.14$, $p=0.068$).

University College Dublin June 2025 - August 2025
Breakthrough Summer Research Scholar (BioSLATE - Biomarker Selection and Synthetic Lethality Analysis for Therapeutic Exploration in high-grade serous ovarian cancer (HGSOC)) - BioSLATE Website | GitHub (BioSLATE) | GitHub (OncoSynth)
- Identified synthetic-lethal targets in HGSOC via significant CNA–CRISPR correlations across >500 tumours, integrating TCGA CNA, DepMap CRISPR and gene-essentiality models; shipped an interactive Streamlit app for target nomination.
- Built OncoSynth, a CrewAI multi-agent CLI automating literature mining, druggability scoring and clinical-trial linkage for translational target discovery.

Royal College of Surgeons Ireland (Final Year Dissertation) January 2024 - March 2024
Visiting Student Researcher - GitHub
- Built a feed-forward neural network on the MeluXina HPC to predict MCF-7 chromatin accessibility from nine epigenetic features (>2.8M bins), reaching 98% accuracy and outperforming published models.
- Ranked features with LOCO and SHAP and benchmarked against the GET foundation model, identifying H3K4me1 as the top sequence-independent enhancer predictor.

PUBLICATIONS

Ogundimu F. & Furney SJ: Deciphering the Non-coding Cancer Genome through Deep Learning and Multiomic Integration (*in revisions - npj Genomic Medicine*)

O'Reilly D, et al.: Biomarkers of Response to Immunotherapy in Advanced Non-small cell lung cancer using Dynamic breath and blood analysis (BRAND) *(in preparation - targeted journal; Journal of Thoracic Oncology)*

Thinnes C. et al.: DrugMap: A genome-scale visualisation of human drug metabolism *(in preparation)*

GRANTS & SCHOLARSHIPS

Government of Ireland (GOI) Postgraduate Scholarship (€136,000, 18% success rate) – Nationally competitive award recognising exceptional research potential.

France Excellence Research Residency Grant (€1,850, French Embassy in Ireland) – Competitively awarded to one of nine awardees nationally; funding a 1-month research residency at Institut Curie (CUBIC, Prof. Nicolas Servant) for CNV-aware chromatin topology analysis and AI-ready regulatory map construction in breast cancer.

RCSI Top-Up Research Grant (€10,000) – Selective supplement to GOI Scholarship supporting CRISPR-based validation of computationally predicted non-coding driver mutations in breast cancer.

Breakthrough Cancer Research Summer Scholarship (€3,000) – National cancer research award supporting 12 hands-on projects in poor-prognosis cancers.

SELECTED PROJECTS & HACKATHON WINS

NCypher — 1st place, Researcher track, Built with Claude: Life Sciences Jul 2026
(Anthropic · Cerebral Valley · Gladstone Institutes; 1 of 299)
GitHub | Demo Video

- Honest MCP triage tool for non-coding cancer variants, chromatin (ChromBPNet), evolutionary constraint and MPRA function, converged with base-level mechanism.

Unravel — 1st place, Fivetran track, Google Cloud Rapid Agent Hackathon Jul 2026
GitHub | Live Link | Demo Video

- Five-agent Gemini 3.1 / Google ADK system that re-scores ACMG pathogenicity when a variant is reclassified and drafts clinician-approved patient recontact (FHIR).

CONFERENCES, PRESENTATIONS & TRAINING

Imperial College London Robotics Summer School (Jul 2026) - Hamlyn Centre for Robotic Surgery.

Invited panel speaker, UCD Women+ in STEM Conference, sponsored by AMD (Apr 2026) - STEM careers, academia vs industry, the future of computational biology.

NVIDIA Deep Learning Institute - Introduction to Deep Learning, Autoencoders & PyTorch (Nov 2025).

Posters: Conway Institute Research Festival, UCD (2025); RCSI Research Day (2026); BRS Research Day, DCU (2025).

ACADEMIC DISTINCTIONS & AWARDS

Conway Institute Research Festival — Poster Commendation (perfect 30/30).

Breakthrough Cancer Research — Best Lay Presentation Award (2025).

BSc. Genetics & Cell Biology — Ranked 2nd, Salutatorian & Dean's Honours List; Faculty Undergraduate Academic Achievement Award; FSH Academic Scholarship.

SKILLS

Programming: Python, R, Bash, Perl, TypeScript

Machine Learning & AI: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, deep learning (CNNs, MLPs), multi-agent systems (CrewAI, Google ADK), RAG, LangChain

Genomics & Data Science: NGS, single-cell ATAC-seq, cfDNA/methylation & bisulfite analysis, survival analysis (Kaplan–Meier, Cox), NMF/NNLS deconvolution, TCGA/DepMap/ENCODE, Docker

Cloud & App: Google Cloud Platform, Vertex AI, FastAPI, Streamlit, React.js, Firebase

Languages: English (native), French (professional working), Irish (conversational)